



Department of Computer Science And Engineering (CSE)

Faculty of Science and Information Technology

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ASSIGNMENT

Course Title : Research Innovation and Technical Writing
Course Code : CSE414
Assignment On : Question Solve

Submitted By	Submitted To
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Comment by Course Teacher:

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Assignment

Ans: to the question no-1(a)

Analysis:

To select the best model based on overall performance we need to compare the performance metrics:

Accuracy, Precision and Recall.

DenseNet169 achieved the best overall performance with 97% accuracy, 94% precision and 95% recall making the highest performing model among the compared models. LW-CNN ranked second by achieving 96% accuracy, 94% precision and 93% recall. Although its accuracy and recall are slightly lower than DenseNet169 it still demonstrates excellent performance. ResNet50 provided satisfactory performance with 85% accuracy, 84% precision and 83% recall outperforming both YOLOv5 and VGG16. YOLOv5 achieved 83% accuracy, 82% precision and 81% recall showing moderate classification performance but remaining below ResNet50. VGG16 showed the lowest performance with 80% accuracy, 77% precision and 78% recall indicating comparatively weaker classification capability.

Final Proposed Model:

Therefore, DenseNet169 is selected as the final proposed model due to its superior overall predictive performance.

Justification:

- (i) It has the highest accuracy (97%).
- (ii) It provides excellent precision (94%), reducing false positive predictions.
- (iii) It has the highest recall (95%), which detects more correct positive cases.
- (iv) It gives the most reliable overall prediction among all models.
- (v) Although it has higher computational cost, its superior prediction performance makes it the best choice when accuracy is the primary objective.

Therefore, DenseNet169 is selected as the final proposed model based on overall predictive performance.

Ans: to the question no-1(b)

For real-world deployment, computational efficiency is very important. A model should have low memory usage, fewer parameters, faster processing and good prediction performance.

Model Analysis:

VGG16 has 25 million parameters and 32 MB model size. It is computationally expensive with relatively low accuracy. ResNet50 has 19 million parameters and

75 MB model size, making it memory-intensive. YOLOv5 has 12 million parameters and 45 MB model size. It is more efficient than VGG16 and ResNet 50. DenseNet169 provides the highest accuracy but requires 45 million parameters and 50 MB model size, which increases computational cost. LW-CNN has only 5 million parameters and 3.2 MB model size while maintaining 96% accuracy, 94% precision and 93% recall.

Most Suitable Model for Deployment:

LW-CNN is the best model for real-world deployment.

Justification:

- (1) It has the smallest model size (3.2 MB)
- (2) It requires the fewest parameters (5M).
- (3) It provides very high accuracy (96%), close to DenseNet169.
- (4) It consumes less memory and processing power.
- (5) It is suitable for deployment on mobile devices, embedded systems and low-resource healthcare environments.
- (6) It offers the best balance between prediction performance and computational efficiency.

P.T.O

Therefore, LW-CNN is the most suitable model for practical real-world deployment because it is lightweight, fast, memory-efficient and highly accurate.

Ans: to the question no-2(a)

(i) Relevant knowledge profile (kp):

• k1-Natural Science:

The system uses medical chest X-Ray images to detect pneumonia. Basic knowledge of human lungs, diseases and medical imaging is required. Therefore, k1 is applicable.

• k2-Mathematics:

Mathematics is required for image preprocessing, data augmentation and performance evaluation. Metrics such as Precision, Recall, F1-Score and Accuracy are mathematical measures. Therefore, k2 is applicable.

• k4-Specialist knowledge:

The project uses deep learning models such as VGG16, ResNet50, DenseNet169 and Lightweight CNN. Knowledge of deep learning and image classification is necessary. Therefore, k4 is applicable.

- k5 - Engineering Design:

Engineers must design an efficient system by selecting the best preprocessing methods, choosing the suitable model and balancing accuracy and computational efficiency. Therefore, k5 is applicable.

- (ii) Relevant Complex Engineering Problem (CEP):

- EP1- Depth of knowledge:

The project requires knowledge of medical imaging, deep learning, image preprocessing and performance evaluation. Therefore, EP1 is applicable.

- EP2- Range of conflicting Requirements:

The system must achieve high prediction accuracy while also maintaining low memory usage and fast preprocessing speed for deployment.

- EP3- Depth of Analysis:

Engineers must analyze different datasets, preprocessing techniques, data augmentation methods, and multiple deep learning models before selecting the final model. Therefore, EP3 is applicable.

P.T.O

- EP7 - Interdependence:

The final system depends on several connected tasks, including data collection, preprocessing, augmentation, model training, evaluation and deployment. Each step affects the overall performance. Therefore, EP7 is applicable.

(iii) Relevant Engineering Activities (EA):

- EA1 - Range of Resources:

The project requires different resources such as medical datasets, deep learning frameworks, image preprocessing techniques, computing devices, and evaluation metrics. Therefore, EA1 is applicable.

- EA2 - Level of Interactions:

The project involves collaboration among AI Engineers, healthcare professionals and researchers to develop and validate the pneumonia detection system. Therefore, EA2 is applicable.

- EA4 - Consequences for Society and Environment:

The developed system will be used in hospitals to

it presents another author's work as their own. Plagiarism is unethical and violates academic integrity.

② Lack of Originality:

The student paraphrased ideas without acknowledging the original authors. Even when using different words, the original source must be cited. Failure to do so reduces the originality of the research.

③ Unethical Use of AI:

The student used an AI writing tool to generate parts of the literature review and discussion without disclosing its use. AI can assist in writing but researchers are responsible for ensuring that the content is accurate, original and properly referenced.

④ Authorship Responsibility:

The student submitted the paper without checking the AI-generated information. Researchers are fully responsible for accuracy and reliability of everything included in their research paper.

support pneumonia diagnosis. Accurate predictions can improve patient care while incorrect predictions may affect treatment decisions. Therefore, EAM is applicable.

Therefore, the most appropriate attributes for this scenario are $KP(K_1, K_2, K_4, K_5)$, $CEP(EP_1, EP_2, EP_3, EP_7)$ and $EA(EA_1, EA_2, EA_4)$ as they match the knowledge requirements, problem complexity and engineering activities described in the scenario.

Ans: to the question no-3(a)

The given scenario presents several ethical issues related to plagiarism, originality, authorship responsibility and the responsible use of AI in research writing. These issues can reduce the quality and creditability of research.

① plagiarism:

The student copied text from published articles without proper citation. This is plagiarism because

⑤ Academic Integrity:

Academic Integrity requires honesty, fairness, transparency and respect for intellectual property. The student's action violate these ethical principles and may reduce the trustworthiness of the research.

⑥ Appropriate Practices to Ensure Academic Integrity:

- ① Always cite all sources using the correct referencing style.
- ② Give proper credit to the original authors when quoting or paraphrasing.
- ③ Use AI only as a supporting tool, not as a replacement for personal research and critical thinking.
- ④ Verify all AI-generated content before including in the paper.
- ⑤ Clearly disclose the use of AI tools if required by the institution or publishers.
- ⑥ Use plagiarism detection software before submission.
- ⑦ Produce original work and follow institutional research ethics guidelines.

The student's action involve plagiarism, lack of originality, irresponsible use of AI, and failure to maintain academic integrity. By following proper citation practices, verifying AI-generated content, maintaining originality and uses of AI in responsibly, researchers can produce ethical, reliable and high-quality research papers.

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